

CS 260 – Foundations of Data Science

Lecture 16 – Principal Component Analysis

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Announcements

HW6 due tonight

HW7 due next Tuesday (will be released tonight)

Midterm 2 – November 19th

- propose to change it to Tuesday November 24th
- Piazza poll

Outline

Principal Components Analysis (PCA)

Singular Value Decomposition (this afternoon)

Example

$$X = \begin{bmatrix} 2 & 2 \\ 1 & 1 \\ -1 & -1 \\ -2 & -2 \end{bmatrix}$$

Do we actually need two numbers to locate every observation?

Project the data

$$X = \begin{bmatrix} 2 & 2 \\ 1 & 1 \\ -1 & -1 \\ -2 & -2 \end{bmatrix}$$

$$\mathbf{v}_1 = \frac{1}{\sqrt{2}}(1,1), \qquad \mathbf{v}_2 = \frac{1}{\sqrt{2}}(1,-1)$$

$$z = \vec{x} \cdot \vec{v}$$

Project the data

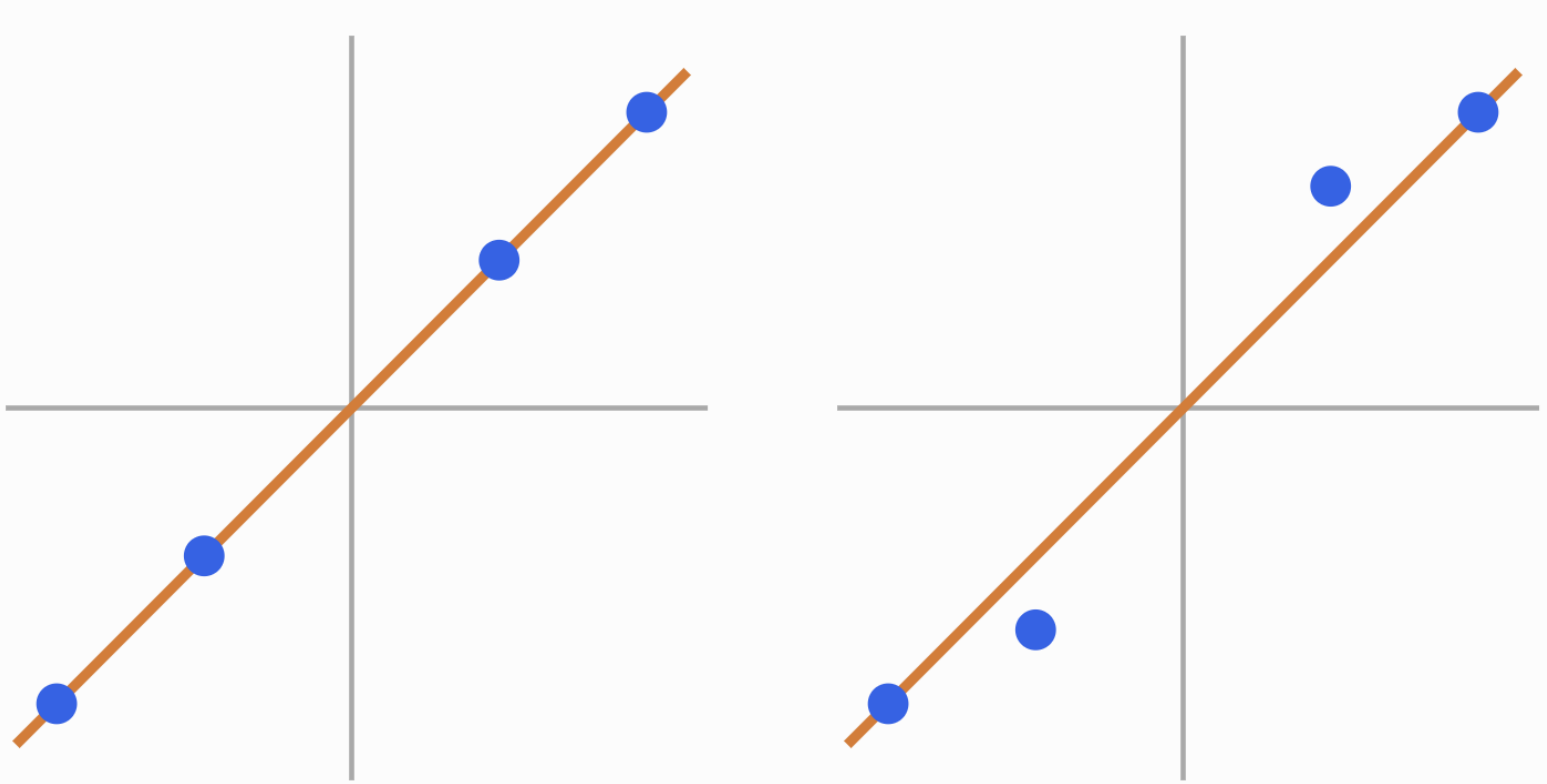
$$X = \begin{bmatrix} 2 & 2 \\ 1 & 1 \\ -1 & -1 \\ -2 & -2 \end{bmatrix}$$

$$\mathbf{v}_1 = \frac{1}{\sqrt{2}}(1,1), \qquad \mathbf{v}_2 = \frac{1}{\sqrt{2}}(1,-1)$$

$$Z = \begin{bmatrix} 2\sqrt{2} & 0 \\ \sqrt{2} & 0 \\ -\sqrt{2} & 0 \\ -2\sqrt{2} & 0 \end{bmatrix}$$

Example 2

$$X_{\text{new}} = \begin{bmatrix} 2 & 2 \\ 1 & 1.5 \\ -1 & -1.5 \\ -2 & -2 \end{bmatrix}$$



Can we still describe every observation perfectly using only one coordinate along the diagonal?

Project the data

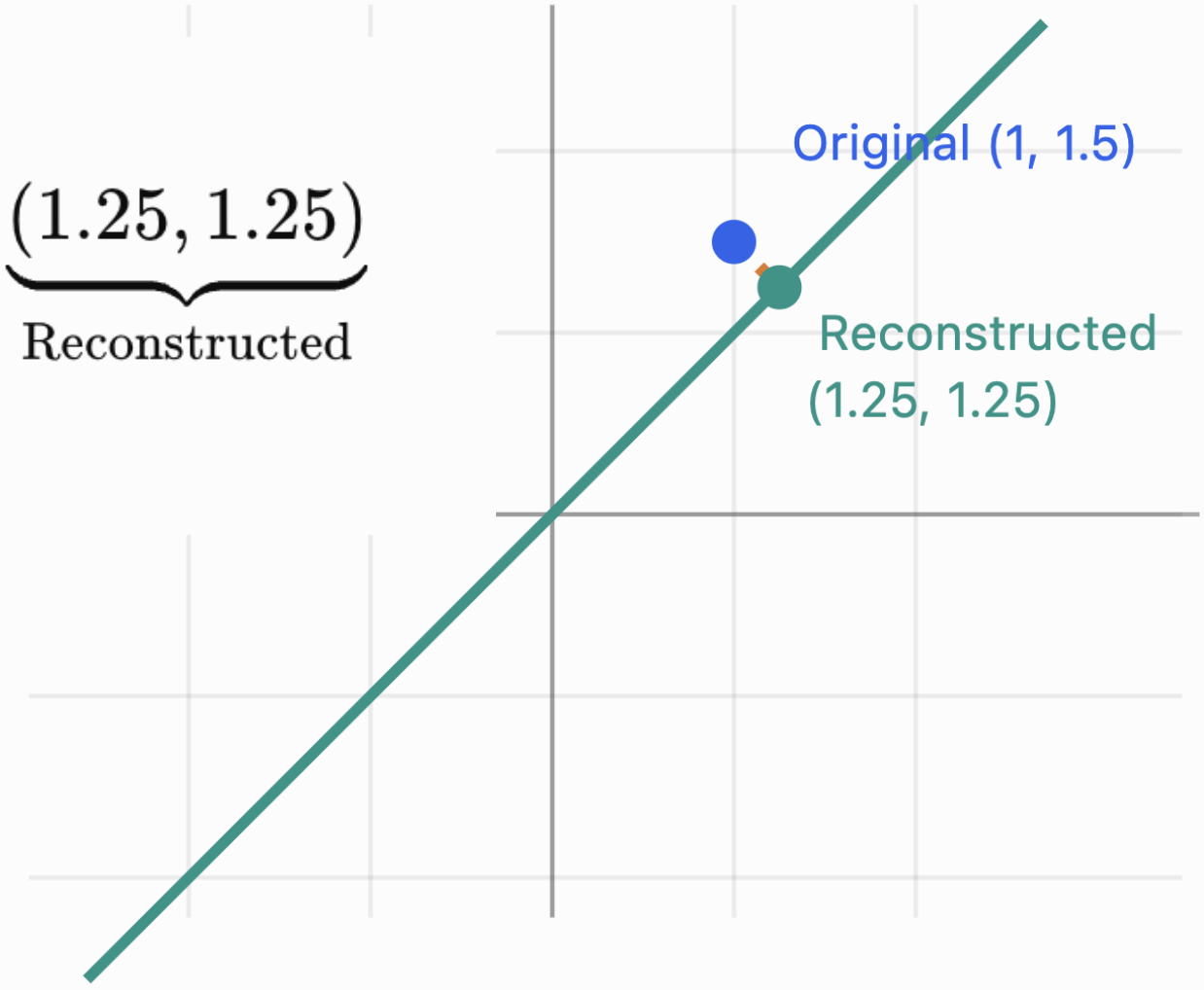
$$X_{\text{new}} = \begin{bmatrix} 2 & 2 \\ 1 & 1.5 \\ -1 & -1.5 \\ -2 & -2 \end{bmatrix}$$

$$\mathbf{v}_1 = \frac{1}{\sqrt{2}}(1,1), \qquad \mathbf{v}_2 = \frac{1}{\sqrt{2}}(1,-1)$$

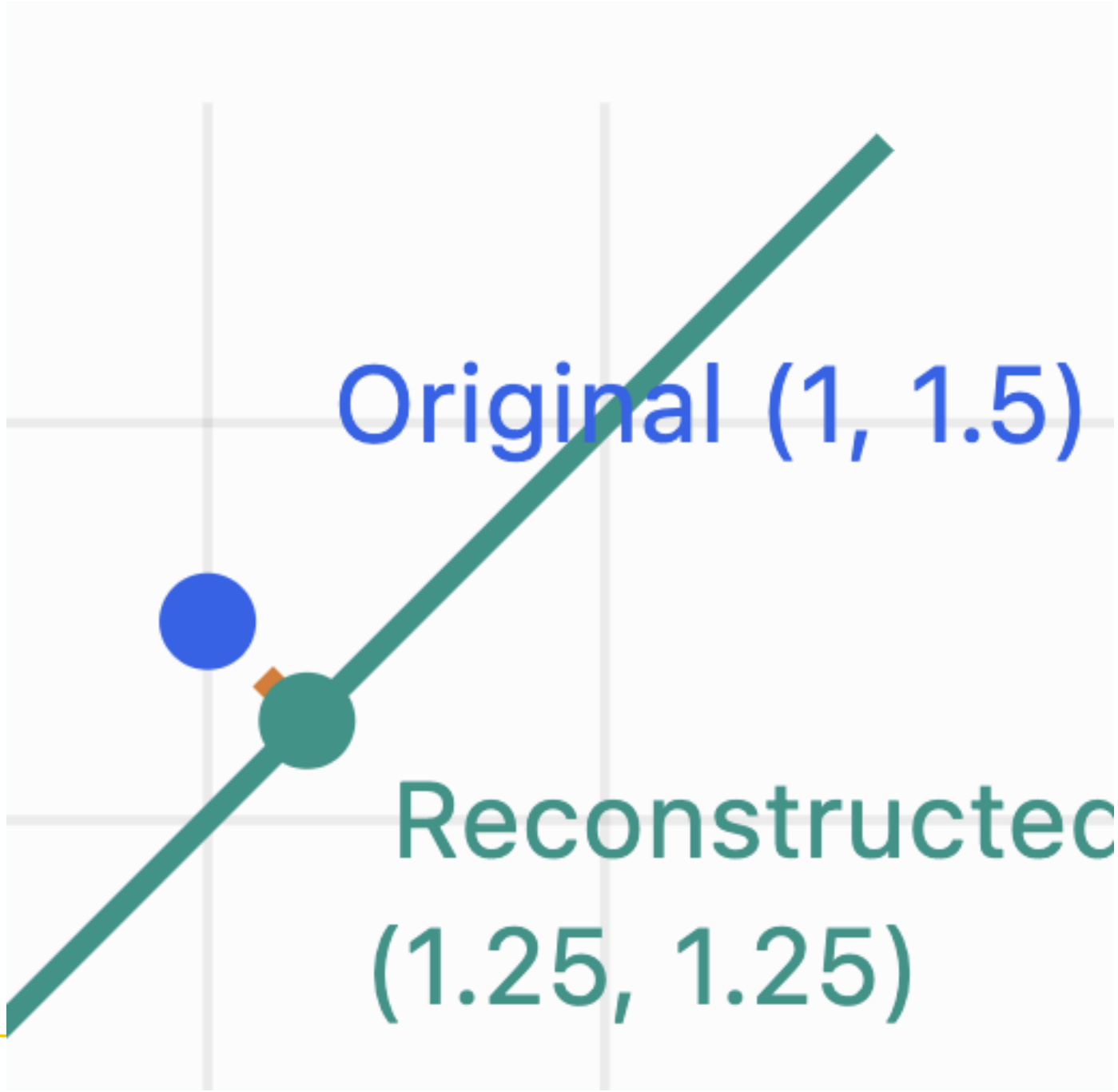
Observation	z_1	z_2
(2,2)	$2\sqrt{2}$	0
(1,1.5)	$2.5/\sqrt{2}$	$-0.5/\sqrt{2}$
(-1,-1.5)	$-2.5/\sqrt{2}$	$0.5/\sqrt{2}$
(-2,-2)	$-2\sqrt{2}$	0

Reconstruct the data

$$\underbrace{(1, 1.5)}_{\text{Original}} \longrightarrow \underbrace{\frac{2.5}{\sqrt{2}}}_{\text{Projected coordinate}} \longrightarrow \underbrace{(1.25, 1.25)}_{\text{Reconstructed}}$$



Reconstruct the data



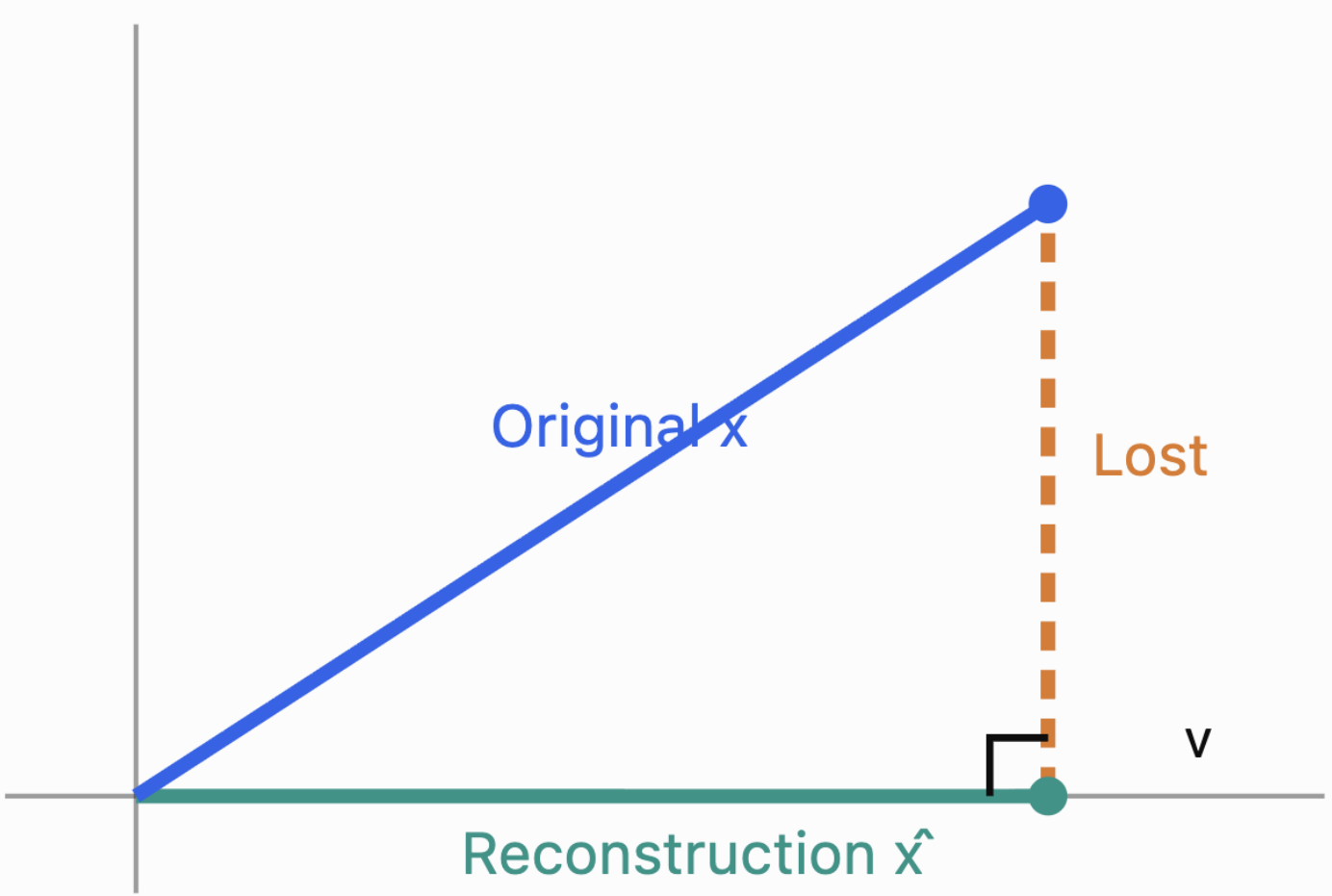
Reconstruct the data

$$||x||^2 = ||\hat{x}||^2 + ||x - \hat{x}||^2$$

$$||x - \hat{x}||^2 = ||x||^2 - ||\hat{x}||^2$$

$$||\hat{x}||^2 = (x \cdot v)^2 = z^2$$

$$||x - \hat{x}||^2 = ||x||^2 - z^2$$



Reconstruct the data for all examples in matrix X

$$\left|\left|\boldsymbol{x} - \hat{\boldsymbol{x}}\right|\right|^2 = \left|\left|\boldsymbol{x}\right|\right|^2 - \boldsymbol{z}^2$$

$$\sum_i \left|\left|\boldsymbol{x_i} - \hat{\boldsymbol{x}}_{\boldsymbol{i}}\right|\right| = \sum_i \left|\left|x_i\right|\right|^2 - \sum_i z_i^2$$

How can we minimize the total reconstruction error?

Amount preserved in Example 1

$$X = \begin{bmatrix} 2 & 2 \\ 1 & 1 \\ -1 & -1 \\ -2 & -2 \end{bmatrix}$$

$$\mathbf{v}_1 = \frac{1}{\sqrt{2}}(1,1),$$

$$Z = \begin{bmatrix} 2\sqrt{2} & 0 \\ \sqrt{2} & 0 \\ -\sqrt{2} & 0 \\ -2\sqrt{2} & 0 \end{bmatrix}$$

$$\sum_i z_i^2 =$$

$$= 8 + 2 + 2 + 8$$

$$= 20$$

Reconstruct the data for all examples in matrix X

$$||x - \hat{x}||^2 = ||x||^2 - z^2$$

$$\sum_i ||x_i - \hat{x}_i|| = \sum_i ||x_i||^2 - \sum_i z_i^2$$

How can we minimize the total reconstruction error?

$$\text{Total squared error} = \text{Fixed} - (n-1)\,\text{Var}(z)$$

Finding direction to maximize variance

$$X = \begin{bmatrix} 2 & 2 \\ 1 & 1 \\ -1 & -1 \\ -2 & -2 \end{bmatrix}$$

$$\mathbf{v}_1 = \frac{1}{\sqrt{2}}(1,1), \qquad \mathbf{v}_2 = \frac{1}{\sqrt{2}}(1,-1)$$

Covariance

How much does two variables move together linearly

Data matrix X
(centered)

$$\begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1d} \\ x_{21} & x_{22} & \cdots & x_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{nd} \end{bmatrix}$$

$n \times d$

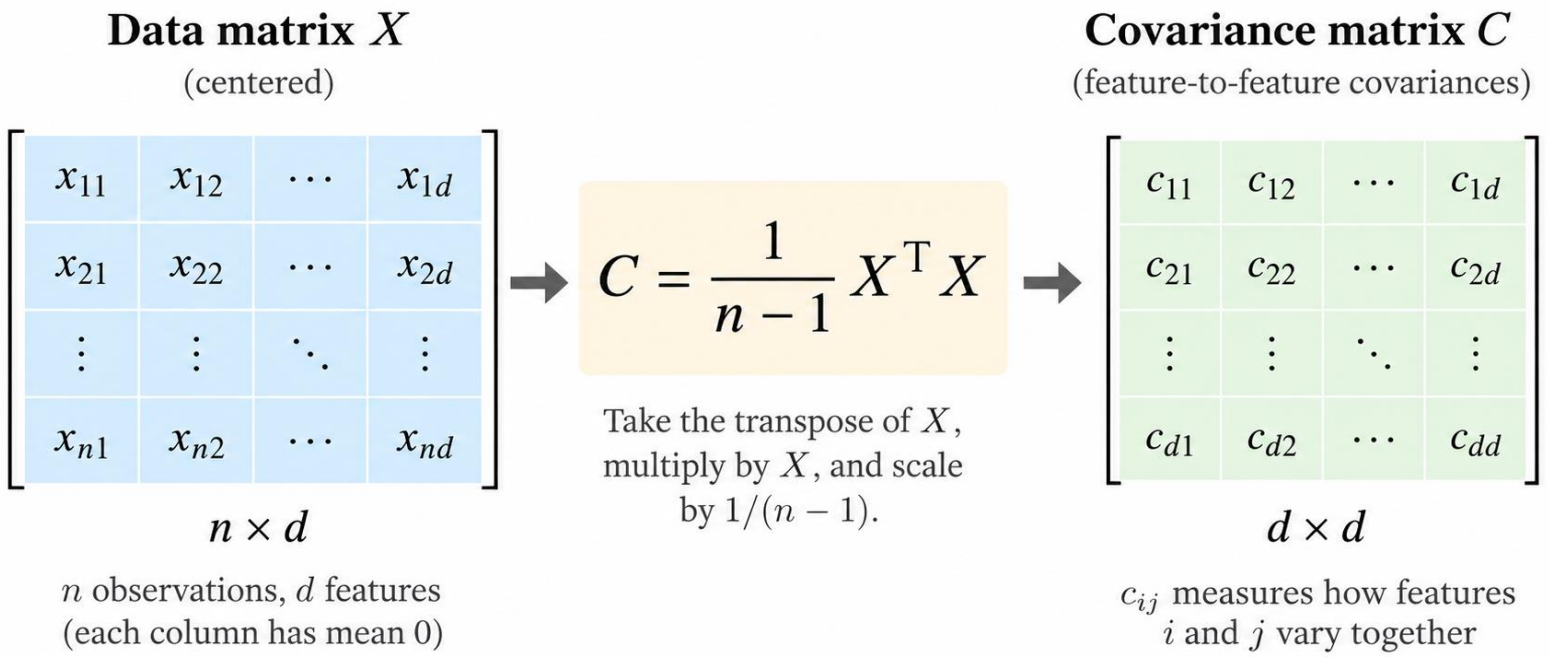
n observations, d features
(each column has mean 0)

Covariance matrix C
feature-to-feature covariances)

$$\begin{bmatrix} c_{11} & c_{12} & \cdots & c_{1d} \\ c_{21} & c_{22} & \cdots & c_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ c_{d1} & c_{d2} & \cdots & c_{dd} \end{bmatrix}$$

Covariance Matrix

How much does two variables move together linearly



Covariance Matrix & Projection & Eigenvectors

PCA

$$X = \begin{bmatrix} 2 & 2 \\ 1 & 1 \\ -1 & -1 \\ -2 & -2 \end{bmatrix}$$

$$C = \frac{1}{3}\begin{bmatrix} 10 & 10 \\ 10 & 10 \end{bmatrix}$$

$$\lambda_1 = \frac{20}{3}, \qquad \mathbf{v}_1 = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\lambda_2 = 0, \qquad \mathbf{v}_2 = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$Z = XV_1$$

$$Z = \begin{bmatrix} 2\sqrt{2} \\ \sqrt{2} \\ -\sqrt{2} \\ -2\sqrt{2} \end{bmatrix}$$

PCA

1. Center our data
2. Computer Covariance Matrix
3. Find eigenvalues and unit eigenvectors
4. Sort eigenvectors by descending eigenvalue
5. Keep only the k eigenvectors
6. Project the observations onto those k directions